

CURRENT POSITION

Traffic App Project Snapshot

Generated July 08, 2026. This is where the project stands after the June 17 data refresh.

Nationwide fatal crashes	All-severity state crashes	County market rows	Tracked source states
186,398 FARS 2019-2023	1,085,643 Utah, Virginia 2019-2024	14,308 county-year FARS plus market context	16 2 automated, 14 pending/review

What We Have

- A working Streamlit dashboard in app.py branded for Astrix Law Traffic Accident Intelligence.
- A reproducible ingestion pipeline in ingest.py that downloads and normalizes government crash, population, and business datasets.
- A local SQLite database plus compressed deployment CSVs, so the app can run locally with full tables or on Streamlit Cloud with a compact bundle.
- Four UI areas: Executive Overview, Accident Trends, Risk Factors, and Geographic Analysis, with state/year filters and source-status disclosure.
- A data provenance memo already exists, though it still uses older Elysium wording while the app now says Astrix Law.

What You Can Tell People

- We can compare fatal crash exposure nationally using NHTSA FARS for 2019-2023.
- We have all-reported crash detail for Utah and Virginia for completed years 2019-2024, including severity, location, weather/light, road context, and commercial indicators.
- We join crash outcomes to county population and Census County Business Patterns for target industries: specialty trade contractors, truck transportation, couriers/messengers, and landscaping services.
- The tool is for aggregate market intelligence, not identifying individual crash victims or proving a specific crash was work-related.

WHERE THINGS ARE

Files, Data, and Next Moves

Project map

File or folder	What it does
app.py	Streamlit UI, navigation, filters, metrics, charts, and map views.
data.py	App-facing loaders. Reads SQLite first, then data/deploy CSV fallbacks.
ingest.py	Full refresh pipeline: downloads raw sources, builds SQLite, exports analysis files.
build_deployment_data.py	Creates compact compressed CSVs for Streamlit Cloud.
eda_data.py	New Python EDA helper with get_data_* functions for each loaded dataset.
data/traffic_data.sqlite	Full local analytical database with ten tables.
data/deploy/	Committed compact app bundle, one compressed CSV per deployed table.
data/exports/	Wider analysis CSVs for R/Python notebooks and ad hoc exploration.
analysis/DATA_DICTIONARY.md	Definitions and EDA caveats for the analysis tables.

Loaded Data

Dataset	Years	Records	Refreshed
NHTSA FARS	2019-2023	186,398	2026-06-17
Census County Business Patterns	2023	8,946	2026-06-17
Census Population Estimates	2020-2024	3,144	2026-06-17
UDOT Utah Crash Locations	2019-2024	354,098	2026-06-17
Virginia Roads Full_Crash	2019-2024	731,545	2026-06-17

Runbook

- Launch the app: streamlit run app.py
- Refresh full local data: python ingest.py
- Rebuild Streamlit deployment files: python build_deployment_data.py
- Start Python EDA: from eda_data import get_data_catalog, get_data_utah_crashes

Open Decisions

- Resolve client/brand naming: older provenance memo says Elysium; current README and app say Astrix Law.
- Decide which pending states are worth source negotiation next; Florida and Texas likely need portal or credential work.
- Choose whether the dashboard should remain market-intelligence only or add recommendation scoring later.